

Assessment No. 01

Total Marks: 20

Submission Deadline:

03-08-2026

Submission Instructions

- Submit the assessment in **printed format only**.
 - Handwritten or soft-copy submissions will **not** be accepted.
 - Attach Java source code after the theory section.
 - Include screenshots of program execution.
 - Proper formatting, commenting, and documentation are mandatory.
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Part A – Theory (10 Marks)

Q1. (2 Marks)

Explain the complete Java Compilation and Execution Process.

Q2. (2 Marks)

Differentiate between:

- JDK
- JVM
- JRE
- JIT Compiler

Draw a simple execution flow diagram.

Q3. (3 Marks)

Discuss the following OOP concepts with suitable Java examples.

- Encapsulation
- Inheritance

- Polymorphism
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Q4. (3 Marks)

Explain Checked and Unchecked Exceptions.

Write the syntax of:

- try
 - catch
 - finally
 - throw
 - throws
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Part B – Programming (10 Marks)

Question 1 (5 Marks)

Develop a **Library Management System** using Java.

Requirements

- Create Book class
- Store Book ID
- Title
- Author
- Availability Status

Implement

- Add Book
- Display Books
- Search Book
- Issue Book
- Return Book

Use

- Constructors
- ArrayList
- Methods

- Exception Handling
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Question 2 (5 Marks)

Create a Java program that

- Reads student information from keyboard.
- Stores it in a text file.
- Reads the file again.
- Displays all stored records.

Concepts Required

- FileWriter
 - BufferedWriter
 - FileReader
 - BufferedReader
 - Exception Handling
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Assessment Learning Outcomes

Students will demonstrate

- Java Fundamentals
- OOP
- Exception Handling
- File Handling
- Problem Solving
- Program Documentation